

action-space to binary decisions ($\mathcal{B} = \mathbb{B}$) to squeeze out the maximum improvement possible through the framework.

Theorem 5.3 (Binary ESA). *For every environment there exists an abstraction and a corresponding surrogate-MDP for its binarized version ($\mathcal{B} = \mathbb{B}$) whose optimal policy is ε -optimal for the true environment. The size of the surrogate-MDP is uniformly bounded for every environment as*

$$|\mathcal{S}| \leq \frac{4[1 - \gamma + \log_2 |\mathcal{A}|]^6}{\gamma^2 \varepsilon^2 (1 - \gamma)^6}$$

Proof. Consider the agent that is interacting with the sequentialized/binarized environment $\check{P}$. By [Theorem 4.6](#), we know that a near-optimal policy of this sequentialized environment is also near-optimal in the original environment. Now, if we use ESA on the binarized problem and get an ε' -optimal policy through the surrogate-MDP by [Theorem 5.2](#), we are sure to be ε -optimal in the original environment P as explained above. Additionally, the size of the state-space is bounded as

$$|\mathcal{S}| \stackrel{\text{Theorem 5.2}}{\leq} \left(\frac{2}{\varepsilon'(1 - \lambda)^3} \right)^2 = \frac{4}{\varepsilon'^2 (1 - \lambda)^6} \quad (22)$$

where λ is the discount factor of the sequentialized problem. Next, we upper bound [Equation \(22\)](#) by using the fact that $\lambda^d = \gamma$. Let $\delta := 1 - \gamma < 1$. So,

$$\begin{aligned} 1 - \lambda &= 1 - (1 - \delta)^{1/d} = 1 - e^{\frac{\ln(1 - \delta)}{d}} \\ &\stackrel{(a)}{\geq} 1 - \frac{1}{1 - \ln(1 - \delta)/d} \stackrel{(b)}{\geq} 1 - \frac{1}{1 + \delta/d} \\ &= \frac{\delta}{d + \delta} = \frac{1 - \gamma}{d + 1 - \gamma} \end{aligned} \quad (23)$$

where (a) holds due to $\frac{1}{e^{-\alpha}} \leq \frac{1}{1 - \alpha}$, (b) is true by using the fact that $\delta < 1$, hence $\ln(1 - \delta) \leq -\delta$. Therefore, using [Equation \(22\)](#), [Equation \(23\)](#), and $\varepsilon' = \lambda^{d-1} \varepsilon \geq \lambda^d \varepsilon = \gamma \varepsilon$ we get,

$$|\mathcal{S}| \leq \frac{4}{\varepsilon'^2 (1 - \lambda)^6} \leq \frac{4(1 - \gamma + d)^6}{\gamma^2 \varepsilon^2 (1 - \gamma)^6} \quad (24)$$

which proves the claim. $\square$

Superficially, it might seem that we have simply replaced the original discount factor with a larger value. But, it is not the case. If we simply scaled the discount factor (without sequentializing the actions) then the resulting bound would indeed deteriorate, see [Theorem 5.2](#), but on the contrary, with sequentialization/binarization and our analysis the bound (dramatically) improves.

Usually in RL the discount factor γ is close to 1. In that case, the bound in [Theorem 5.3](#) can be tightened further as:

$$|\mathcal{S}| \lesssim \frac{4[\log_2 |\mathcal{A}|]^6}{\varepsilon^2 (1 - \gamma)^6} \quad (25)$$

which agrees with the bound in [Theorem 5.2](#) for the case when $|\mathcal{A}| = 2$, i.e. when the original problem already has a binary action-space.

6 Conclusion & Outlook

This work contributes to the study of the GRL problem. We have provided a reduction to handle large state and action spaces by sequentializing the decision-making process. This helped us improve the upper bound on the number of states in ESA from an exponential dependency in $|\mathcal{A}|$ to logarithmic. The gain is *double exponential* in terms of the action-space dependence at no other cost. Our result carries a broader impact on the implementation of *general* RL agents¹². The required storage for such agents, which have access to a non-MDP, approximate Q-uniform abstraction, can be reasonably bounded which only scales logarithmically in the size of the action-space.

We conclude the paper with some future research directions. This work analyses the case when the agent has a fixed aggregation map. [?] provides an outline for a learning algorithm to learn such abstractions which can be combined with our sequentialization framework. Another direction, which we also did not touch in this work, is to explore the connection, if any, between the surrogate-MDPs of a map on the original environment, and its extension on the sequentialized problem. By lifting the small binary ESA map, say ψ , back to $\mathcal{H}$, one obtains a small map directly on $\mathcal{H}$, say ϕ . While ψ used sequentialization/binarization for the construction of ϕ , the map ϕ can be used without further referencing to sequentialization. This suggests that a bound logarithmic in $|\mathcal{A}|$ should be possible without a detour through the sequentialization. This deserves further investigation.

We sequentialize the action-space through an *arbitrary* coding scheme C , so the main result does not depend on this choice. Sometimes, it is possible that the action-space may allow natural sequentialization, e.g. in a video game controller the macro action might be a binary vector where the first bit might represent the left/right direction, the second bit indicates up/down, and so on. The exact nature of these binary decisions depends on the domain which is reflected by the choice of encoding C . Sequentialization was our path to double-exponentially improve that bound. Whether there are more direct/natural aggregations with the same bound is an open problem. Moreover, if the agent is learning an abstraction through interaction, the choice of these functions may become critical. This paper focused on rigorously formalizing and proving the main improvement result. One can also try to empirically show the effectiveness of our improved upper bound. To do this, we need a problem domain where ESA requires more states than the sequentialized/binarized version of it. But a point of caution is that the upper bound still scales badly in terms of γ and ε . Any reasonable value of these parameters would imply a huge upper bound. Even with Markovian abstractions, a cubic dependency on the discount factor is the best achievable. We considered a general underlying process and non-Markovian abstractions, and dramatically improved the previously best bound $(1 - \gamma)^{-3|\mathcal{A}|}$ to $(1 - \gamma)^{-3 \cdot 2}$. Indeed it would be interesting to see whether this can be further improved to the optimal $(1 - \gamma)^{-3}$ rate.

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¹²The general (or strong) agents are designed to work with a wide range of environments [?].